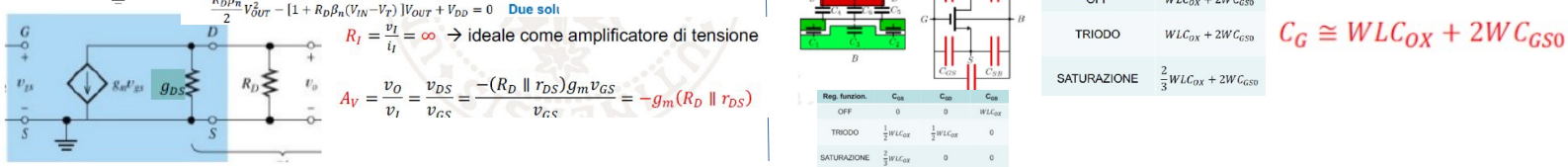
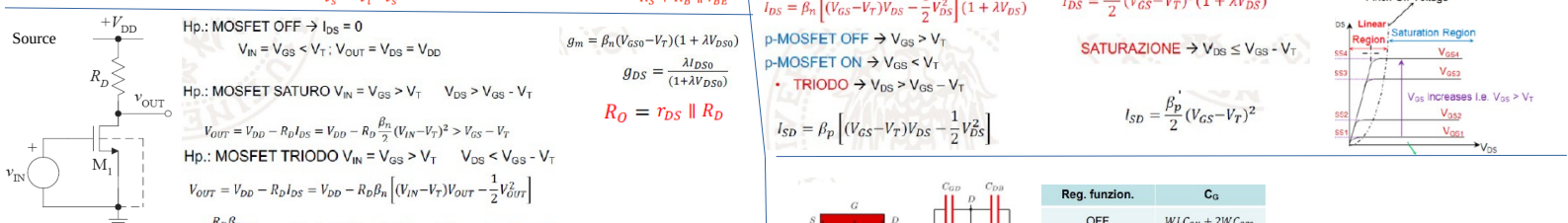
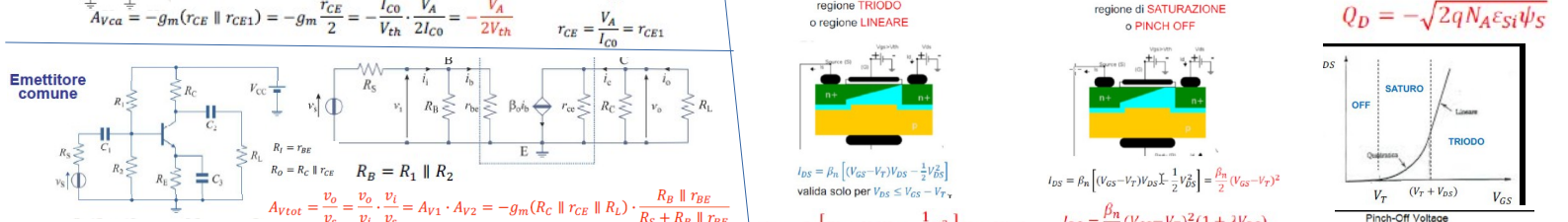
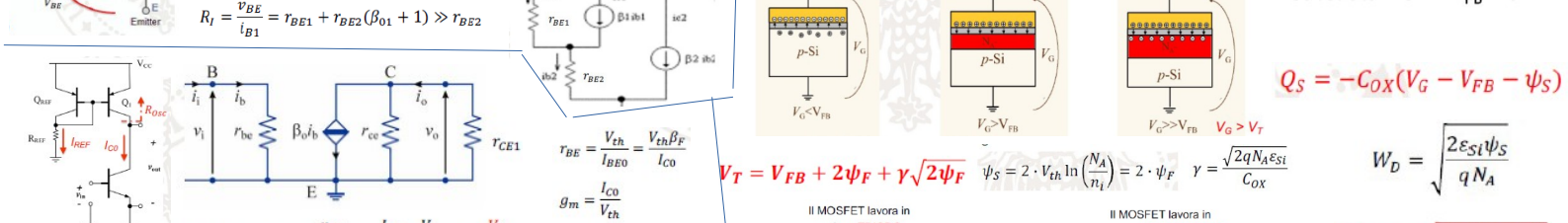
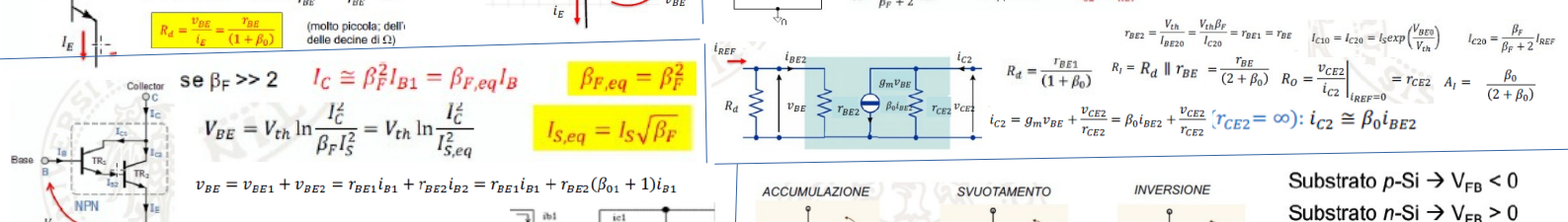
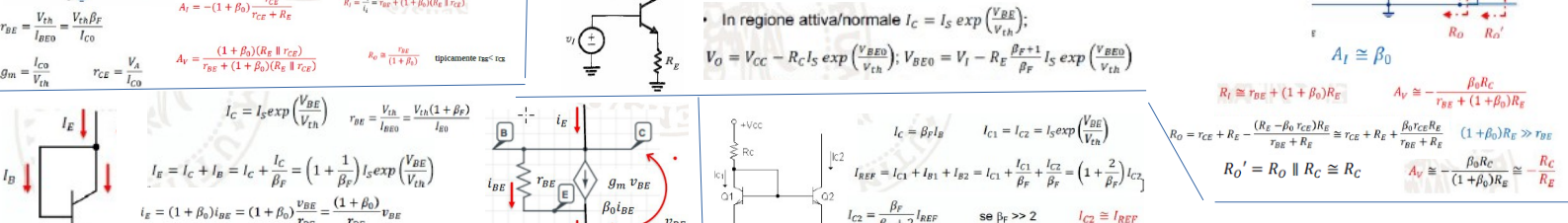
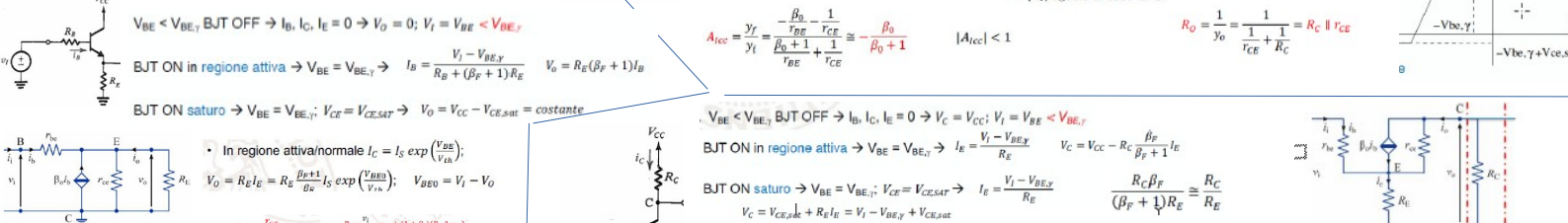
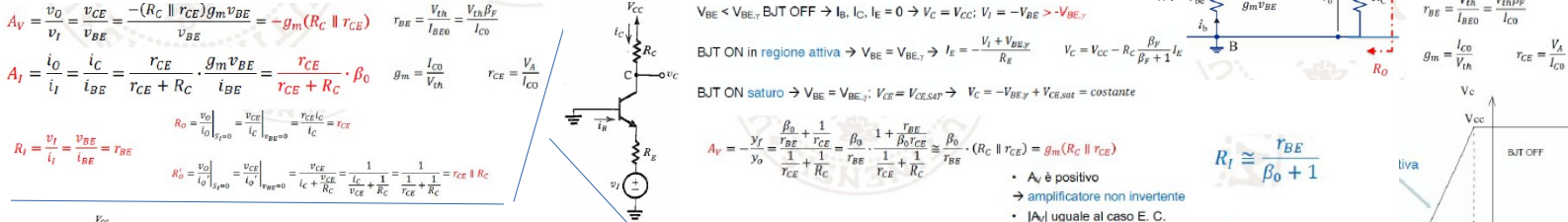
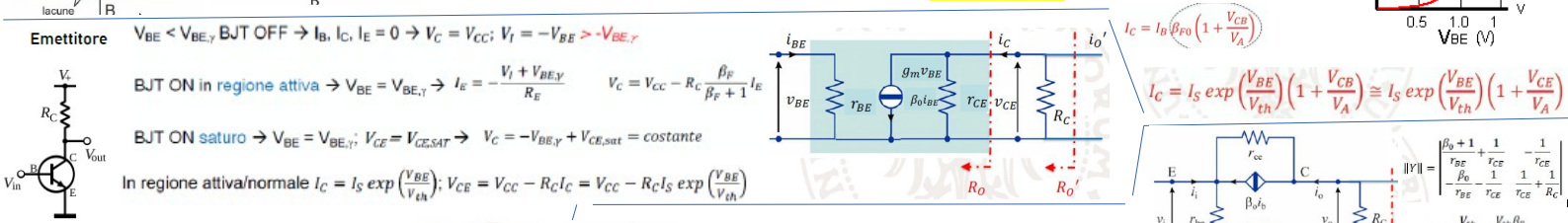
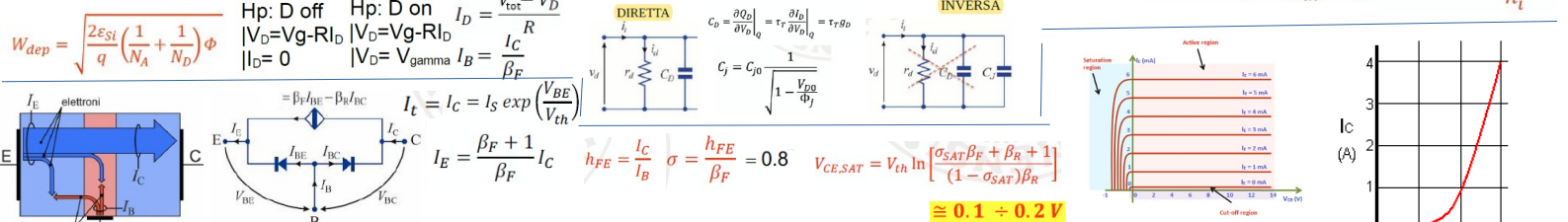
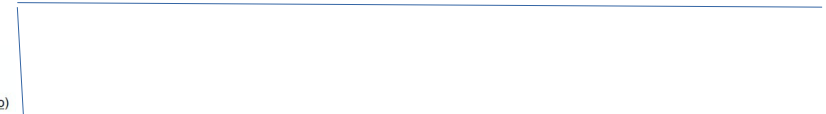
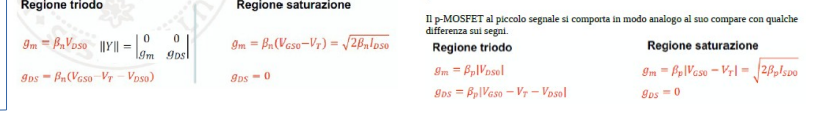
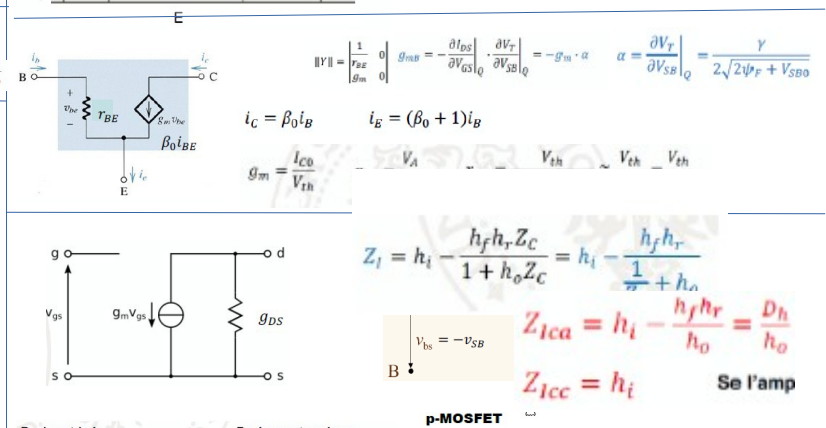
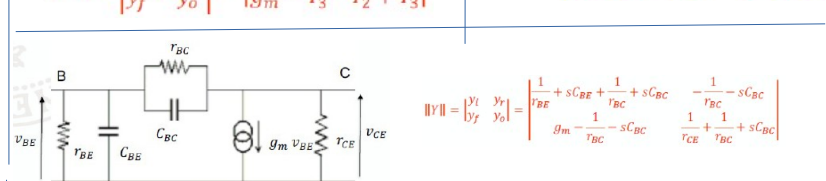
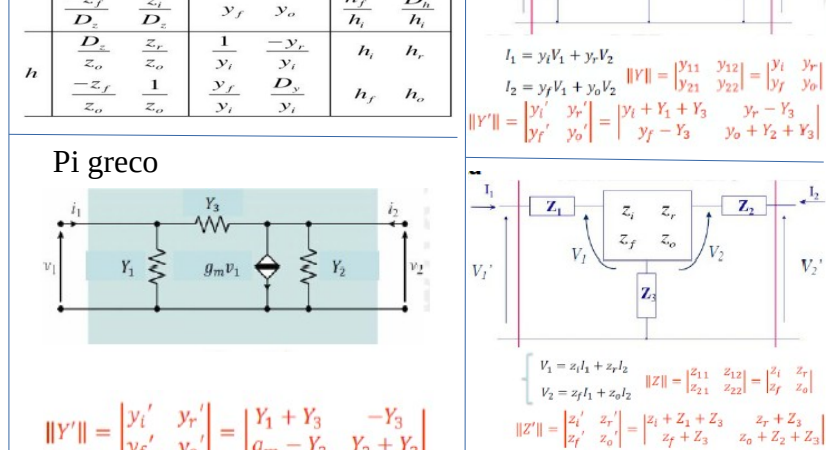
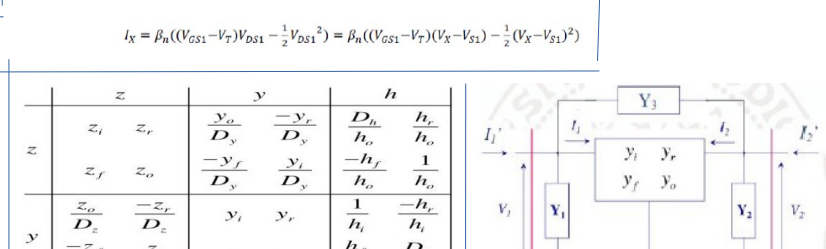
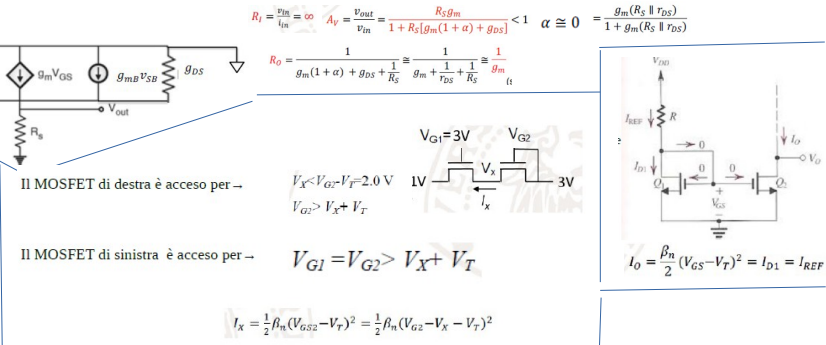


$$J_{tot} = J_{tot,n} + J_{tot,p} = q(\mu_{n,n} + \mu_{p,p}) E + q D n \nabla n - q D p \nabla p \quad V_D = V_{th} \ln \left(\frac{I_D}{I_S} + 1 \right) \quad D_n / \mu_n = D_p / \mu_p = k_B T / q = V_{th} = 25.6 \text{ mV} \quad I_D = A \cdot J_S \cdot \left[\exp \left(\frac{V_D}{V_{th}} \right) - 1 \right] \quad \Phi_j = V_{th} \ln \frac{N_A N_D}{n_i^2}$$





	Emettitore comune	Collettore comune	Base comune
A_v	$-\frac{\beta_o (R_C \parallel r_{ce})}{r_{be}}$	$\frac{(\beta_o + 1)(R_E \parallel r_{ce})}{r_{be} + (\beta_o + 1)(R_E \parallel r_{ce})}$	$\frac{\beta_o}{r_{be}} (R_C \parallel r_{ce})$
A_i	$\beta_o \frac{r_{ce}}{r_{ce} + R_C}$	$-(\beta_o + 1) \frac{r_{ce}}{r_{ce} + R_E}$	$-\frac{\beta_o r_{ce}}{R_C + (\beta_o + 1)r_{ce}}$
R_{in}	r_{be}	$r_{be} + (\beta_o + 1)(R_E \parallel r_{ce})$	$\frac{r_{BE}}{\beta_o + 1}$
R_{out}	$R_C \parallel r_{ce}$	$\frac{r_{BE}}{\beta_o + 1}$	$R_C \parallel r_{ce}$

